

Hoang Vu

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vunas | Portfolio



Education

Saigon University

Expected 2027

B.S. in Software Engineering, Faculty of Information Technology | GPA: 3.11/4.0 Ho Chi Minh City, Vietnam

- Current Status:** 3rd-Year Student (Seeking a full-time Internship with opportunities for long-term employment).
- Core Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems, Software Architecture, Web Development.

Technical Skills

Languages: TypeScript, JavaScript, Python, PHP, SQL, HTML/CSS

Frontend: React.js, Next.js, Tailwind CSS, Blade Templates

Backend: Node.js, NestJS, FastAPI, Laravel

Databases: PostgreSQL, MySQL

Tools & Architecture: Git/GitHub, Docker, Local LLMs (Ollama), RAG, REST APIs, JWT

Selected Projects

Warehouse Management System (WMS) | Live Demo | GitHub

Jan 2026 – June 2026

Tech: Laravel, PHP, PostgreSQL, Tailwind

- Developed a comprehensive warehouse management system supporting centralized inventory tracking across a dynamic location hierarchy (zones, racks, shelves, and bins).
- Designed a PostgreSQL schema to manage products, batches, suppliers, and internal stock movements efficiently.
- Developed dynamic inbound and outbound warehouse processes with a hybrid FEFO/FIFO stock allocation algorithm, improving inventory rotation and shelf-life management.
- Engineered an automated inventory monitoring system using Laravel Jobs and cron scheduling to dispatch smart email alerts for low-stock and expiring batches.

SecureSign (Digital Signature System) | Live Demo | GitHub

Apr 2026 – Present

Tech: FastAPI, PostgreSQL, Supabase, SQLAlchemy, Tailwind

- Built a full-stack web tool for uploading PDF files, generating embedded digital signatures (PAdES standard), and validating document authenticity.
- Applied RSA cryptography algorithm for secure key generation to prevent document tampering.
- Utilized FastAPI and SQLAlchemy to handle user signing flows, integrated with Supabase for robust cloud storage and deployment compatibility.
- Implemented event logging via PostgreSQL JSONB to track user actions and cryptographic audit trails.
- Implemented CI/CD pipelines and ensured backend reliability with 81% test coverage using PyTest.

POS Management System | Live Demo | GitHub

Mar 2025 – May 2025

Tech: ReactJS, Vite, NestJS, TypeORM, MySQL, JWT, Tailwind

- Developed a full-stack Point of Sale (POS) management system tailored for restaurants to streamline sales, table booking, and order processing.
- Built a responsive frontend using ReactJS, Vite, and TailwindCSS, incorporating dynamic dashboard analytics, customer feedback loops, and QR code integration for table ordering.
- Automated invoice generation, and state-driven table status management (Active, Locked, In-use, Reserved).

SmartDoc AI (Local Document RAG) | GitHub

Mar 2026 – May 2026

Tech: Python, LangChain, FAISS, Ollama

- Developed a local Retrieval-Augmented Generation (RAG) tool to parse, analyze, and query offline documents.
- Integrated local Large Language Models via Ollama and FAISS for efficient vector storage and accurate text retrieval.

Experience

Academic Project Collaborator

Jan 2024 – Present

Saigon University

Ho Chi Minh City, Vietnam

- Collaborated in teams of 3-5 members to design and develop full-stack web applications following Agile workflows.
- Applied Object-Oriented Programming (OOP) principles to write readable, maintainable, and structured source code.
- Utilized Git and GitHub for version control and actively participated in code reviews to ensure code quality.